Junghyun (Joanne) Yun

Madison, WI | jyun45@wisc.edu | +1 608-895-5470
linkedin.com/in/joannejyun | github.com/joannemiki57

Research interests: Machine learning for manufacturing, acoustic process monitoring, visual defect inspection, and generalization across heterogeneous production data.

EDUCATION

University of Wisconsin–Madison Sep 2025 – Present
M.S., Industrial & Systems Engineering; expected May 2027
Visiting International Student Program, Industrial & Systems Engineering

Ajou University, Suwon, South Korea Mar 2023 – Aug 2026
B.S., Industrial Engineering | **Class rank: 1/139** | **GPA: 4.26/4.5**

PUBLICATIONS

[1] Zhang, P., **Yun, J.**, and Qin, H. "**Federated Learning-Based Regression Framework to Predict Printed Line Width in Electrohydrodynamic Inkjet Printing.**" 54th SME North American Manufacturing Research Conference (NAMRC 54), 2026. **Accepted; second author.**

[2] Ouidadi, H., Alharbi, K., Zhang, P., **Yun, J.**, and Qin, H. "**Deep Learning-Based Perforation Defect Detection and Classification for Automated Industrial Inspection: An Industry Case Study.**" **In preparation.**

RESEARCH EXPERIENCE

University of Wisconsin–Madison | FEAP Lab Madison, WI
Flexible Electronics & Additive Printing Lab | Advisor: Prof. Hantang Qin

Research Assistant, Sep 2026 – Present; **Undergraduate Researcher**, Sep 2025 – May 2026

Acoustic monitoring for laser powder bed fusion

- Developed a **Python** pipeline to isolate printing layers from LPBF chamber audio, supporting a funded project on in-situ material-property prediction for EUROFER97 steel.
- Combined acoustic equalization, harmonic–percussive separation, and **log-mel template matching** to detect recoater events; identified layer boundaries from laser-restart energy changes.
- Segmented a **5.2-hour recording into 209 layer segments** and its shorter companion channel into **135**; exported WAV clips and an Excel timing index, with manual spot checks of boundaries.

Cross-system line-width prediction in inkjet printing [1]

- Built individual **XGBoost** regressors and data-processing pipelines for **8 printing-system clients** in a study reporting **1,185 experimental runs**, establishing the baseline for a federated learning framework.
- Showed that a strong within-client model (**$R^2 = 0.983$**) yielded **negative R^2 on four clients** when transferred, revealing cross-system generalization failure and motivating the federated comparison.

Industrial surface-defect detection [2]

- Co-led defect-inspection research with a postdoctoral researcher; designed an approximately **900-image annotation strategy** for **25,000 unlabeled production images** and co-wrote sampling and pseudo-labeling methods for [2].
- Evaluated an adaptive-thresholding pseudo-labeling pipeline through component ablations and six-parameter sensitivity analysis. Located **93.4% of 852 regions across 213 images**, with **18 false boxes**, under a center-based localization metric.
- Reconstructed a Faster R-CNN training pipeline, reaching **85.0% AP / 97.6% AP50** on 178 validation images. Research received **3rd of 33 teams, IISE QCRE Data Challenge**, and **2nd Place, IEOM Senior Capstone Design Competition** (2026).

RESEARCH EXPERIENCE CONTINUED

Purdue University | Research Intern Jun – Aug 2026

Laboratory for Advanced Multiscale Manufacturing | Prof. Martin Jun | West Lafayette, IN

Acoustic process monitoring in directed energy deposition

- Analyzed a **six-run subsystem-isolation study** to distinguish laser, powder-feed, and motion contributions in manufacturing audio. Aligned audio, video, and machine logs using oxygen readings, with a **0.17 s alignment residual** in the analyzed recordings.
- Mapped G-code operations to machine-log timestamps and segmented audio by contour, infill, traverse, and layer transition, enabling matched laser-on/off comparisons of the same operations.
- Found that apparent broadband laser events also occurred in laser-off controls; event averaging over **45 laser-on edges** revealed a **7–8.5 kHz transient** absent from the corresponding control, identifying a candidate process signature for further study.

Geometry-based surround-view camera calibration

- Proposed and implemented a targetless calibration method using known vehicle dimensions and image boundaries to refine camera roll, aiming to narrow the image-search region for online road calibration.
- Implemented gradient-based boundary extraction and robust optimization in **Python/OpenCV**; compared calibration methods and investigated systematic bias and poorly conditioned views.
- Contributed to a **C++/OpenCV** surround-view simulator and implemented interactive camera-pose switching; ported a road-scene selection pipeline using feature tracking and geometric outlier rejection.

Ajou University | Undergraduate Research Intern Mar 2023 – Aug 2025

New Industrial Convergence R&D Lab | Advisor: Prof. Jooyeon Lee | Suwon, South Korea

- Developed API-based telemetry acquisition for solar-panel digital-twin monitoring and designed an anomaly-detection model for a project with **Korea Midland Power and Kernelog**.
- Led a microfluidic sweat-sensing project combining PDMS fabrication, colorimetric pH sensing, and image-based analysis; received **2nd Prize, Witeck field-based technical problem-solving project** (2024).

TEACHING EXPERIENCE

Reader/Grader, ISyE 320 | UW–Madison Sep 2026 – Present

Simulation and Probabilistic Modeling, Prof. Qiaomin Xie. Sole grader for **approximately 80 students**, with responsibility for homework and exam grading and written feedback.

Teaching Support, KIAT Data Analytics Program | Purdue Jun – Aug 2026

Led hands-on Python labs, held office hours, and debugged student code in a four-week program serving **90+ visiting scholars** across **20 sessions / approximately 60 contact hours**.

SELECTED RESEARCH COMMUNICATION AND LEADERSHIP

Co-Entrepreneurial Lead, NSF I-Corps BOOST, Summer 2026. Conducted approximately 10 customer-discovery interviews to investigate knowledge-management needs in nuclear operations.

Conference presentation: "Bicycle Lift Installation Optimization based on Spatial Information," Korean Society of Systems Engineering, May 2025. Used QGIS and public data for infrastructure siting.

TECHNICAL SKILLS

Programming and data: Python, C++, SQL, NumPy, pandas, SciPy, Git, CMake.

Machine learning and vision: PyTorch, scikit-learn, XGBoost, Detectron2, OpenCV; object detection, cross-client evaluation, camera calibration, and component ablation.

Signal processing: STFT, log-mel features, acoustic equalization, template matching, event averaging, and synchronization of audio with video and machine logs.